A065-v0.3.7 | Finite-Core Inference Falsifier

Status: SYNTHETIC METHOD CHECK. No independent physical PKLSA core source available.
Precommit SHA256: 486414f5da90661ca5e13123f44290a3651ccac26ab71d260591656947f88429

1. Hypothesis and observables

H0: a centerline and circulation alone do not fix a physical vortex core or a universal numerical smoothing epsilon.
Gamma and cross-section enstrophy Z are training observables. Held-out radial swirl and circular Euler pressure-gradient controls are never used to infer epsilon.

2. Analytical mapping

Gaussian omega(r) = Gamma / (pi*a^2) * exp(-r^2/a^2). Z = Gamma^2/(2*pi*a^2); a = Gamma/sqrt(2*pi*Z).
Rankine omega(r) = Gamma/(pi*a^2) inside core, zero outside. Z = Gamma^2/(pi*a^2); a = Gamma/sqrt(pi*Z).
Rosenhead swirl: v_R = Gamma*r/(2*pi*(r^2+epsilon^2)). Matching near-axis v/r yields epsilon=a; matching half-circulation yields a different epsilon.
Radial pressure gradient for stationary azimuthal Euler control: dp/dr = rho*v_theta^2/r.

3. Gate results

G22: PASS
G23: PASS
G24: FAIL
G25: PASS
G26: PASS
G27: UNRESOLVED_NO_QUALIFIED_PHYSICAL_CORE_PROVIDER
G24 FAIL is an expected rejection of universal Rosenhead representation, not a failure of source recovery.

4. Independent held-out predictions

Rows: id / derived a / epsilon by half circulation / Rosenhead swirl error / pressure error / wrong-profile error
gaussian_1: 0.0750000 / 0.0624416 / 16.714% / 32.016% / 17.365%
gaussian_2: 0.1300000 / 0.1082321 / 17.538% / 31.634% / 18.562%
gaussian_3: 0.2100000 / 0.1748365 / 18.838% / 31.614% / 20.338%
rankine_1: 0.0750000 / 0.0530330 / 31.291% / 61.696% / 17.004%
rankine_2: 0.1300000 / 0.0919239 / 32.556% / 61.602% / 18.756%
rankine_3: 0.2100000 / 0.1484924 / 34.556% / 61.627% / 20.194%

5. Physical closure blocker

Original knot XYZ specifies filament centerline geometry only, not transverse vorticity, cross-section enstrophy, SI dimensions or an independently validated Euler evolution.
No identification of a physical SST rc; no pressure-force, surface, Casimir or gravity claim.
The synthetic pressure gradient is algebraically generated from the same synthetic swirl field and is not an independent measurement.

6. Reproduction

Run: run_all.cmd CORE; run_all.cmd SELFTEST; run_all.cmd ALL with frozen SST framework v1.0.7 installed.

G26 C++17 is kernel-only; no independent full PDE integration.